

Omega-3 fatty acids for nutrition and medicine: considering microalgae oil as a vegetarian source of EPA and DHA.

Long-chain EPA/DHA omega-3 fatty acid supplementation can be co-preventative and co-therapeutic. Current research suggests increasing accumulated long chain omega-3s for health benefits and as natural medicine in several major diseases. But many believe plant omega-3 sources are nutritionally and therapeutically equivalent to the EPA/DHA omega-3 in fish oil. Although healthy, precursor ALA bio-conversion to EPA is inefficient and production of DHA is nearly absent, limiting the protective value of ALA supplementation from flax-oil, for example. Along with pollutants certain fish acquire high levels of EPA/DHA as predatory species. However, the origin of EPA/DHA in aquatic ecosystems is algae. Certain microalgae produce high levels of EPA or DHA. Now, organically produced DHA-rich microalgae oil is available. Clinical trials with DHA-rich oil indicate comparable efficacies to fish oil for protection from cardiovascular risk factors by lowering plasma triglycerides and oxidative stress. This review discusses 1) omega-3 fatty acids in nutrition and medicine; 2) omega-3s in physiology and gene regulation; 3) possible protective mechanisms of EPA/DHA in major diseases such as coronary heart disease, atherosclerosis, cancer and type 2 diabetes; 4) EPA and DHA requirements considering fish oil safety; and 5) microalgae EPA and DHA-rich oils and recent clinical results.